

Lecture 4 Tutorial — Parameter Estimation & Hypothesis Testing

Q1. Core Concepts (Fill in the blanks)

Complete the definitions.

1. A _____ estimate is a single value used to estimate a population parameter.
2. The _____ level is (for example, 0.95).
3. For the standard normal distribution, a _____ value is a z score on the borderline separating significantly low/high values.
4. A _____ interval gives a range of plausible values for a population parameter.

Q2. Confidence Interval Interpretation (Multiple Choice)

Choose the best interpretation of a 95% confidence interval for a population proportion p .

- A. There is a 95% chance that the true value of p is in this specific interval.
- B. We are 95% confident that the interval produced by this method contains the true value of p .
- C. 95% of sample proportions must fall inside this interval.
- D. The population proportion changes with probability 0.95.

Q3. Confidence Level and Tails (Calculation Fill-in)

For a 95% confidence level, complete the values:

1. $\alpha =$ _____
2. $\alpha/2 =$ _____
3. $1 - \alpha/2 =$ _____

The cumulative area to the left of the positive critical z value is _____.

Q4. Determining Sample Size (Short Answer)

The lecture lists three ingredients for determining sample size (for proportion estimation).

Write all three items exactly in statistical terms (short phrases are sufficient).

Q5. Estimating a Population Mean (Rule Selection)

For each case, select the distribution/test statistic form suggested by the lecture when estimating a population mean.

1. $n = 45 \rightarrow$ use _____ (Normal / Student t)
2. $n = 18 \rightarrow$ use _____ (Normal / Student t)

Complete the test statistic shown in the lecture:

$$t = \frac{\bar{x} - \mu}{\text{_____}/\sqrt{n}}$$

Q6. Estimating a Population Variance (Formula + Interpretation)

Complete the chi-square statistic for population variance:

$$\chi^2 = \frac{(n-1)s^2}{\text{_____}}$$

In this formula, s^2 is the _____ variance (sample / population), and σ^2 is the _____ variance (sample / population).

Q7. Null and Alternative Hypotheses (Construction)

Scenario (from lecture theme): compare plant growth under Fertilizer A and Fertilizer B.

Construct hypotheses for the claim: "There is a difference in average plant growth between the two fertilizers."

H_0 : _____

H_a : _____

Then identify the test type: _____ (left-tailed / right-tailed / two-tailed)

Q8. Significance Level and Error Interpretation (True / False + Correction)

Write T (True) or F. If false, rewrite the statement correctly in one line.

1. The significance level is the probability of rejecting H_0 when H_0 is true.
2. A larger α generally makes rejection of H_0 harder.
3. The parameter tested is always a mean.

Q9. Parameter-to-Distribution Mapping (Matching)

Match the parameter being tested to the sampling distribution indicated in the lecture.

Parameters

- A. Mean
- B. Variance
- C. Proportion

Distributions

1. Normal distribution
2. Chi-square distribution
3. Student distribution (t)

Write your answers as: A-____, B-____, C-____.

Q10. Test Type, Direction, and Critical Region (Structured Multiple Choice)

For each alternative hypothesis, choose the correct test type.

1. $H_a : \mu \neq \mu_0$
 - A. Left-tailed
 - B. Right-tailed
 - C. Two-tailed
2. $H_a : \mu < \mu_0$
 - A. Left-tailed
 - B. Right-tailed
 - C. Two-tailed
3. $H_a : \mu > \mu_0$
 - A. Left-tailed
 - B. Right-tailed
 - C. Two-tailed
4. The lecture states that different test types lead to different selected _____ values.
 - A. sample means
 - B. z scores (critical values)
 - C. variances
 - D. confidence levels

Q11. Decision Rule and Conclusion Wording (Applied Reasoning)

Use the lecture decision rule: If $P\text{-value} \leq \alpha$, reject H_0 .

For each case, decide whether to Reject or Fail to reject H_0 .

1. $P = 0.031$, $\alpha = 0.05$
2. $P = 0.120$, $\alpha = 0.05$
3. $P = 0.010$, $\alpha = 0.01$
4. $P = 0.0101$, $\alpha = 0.01$

Then complete the standard reporting phrase (one line):

“If we fail to reject H_0 , there is _____ evidence to support / warrant rejection of the claim.”

Q12. Two Samples, ANOVA, and Non-parametric Tests (Integrated Question)

Part A — Two-sample proportion hypotheses (Fill in the blanks)

Complete the equivalent null and alternative forms shown in the lecture:

$$H_0 : p_1 = p_2 \iff H_0 : p_1 - p_2 = \underline{\hspace{2cm}}$$

$$H_1 : p_1 \neq p_2 \iff H_a : p_1 - p_2 \neq \underline{\hspace{2cm}}$$

Part B — One-way ANOVA (Multiple Choice)

One-way ANOVA is used to test:

- A. Equality of two population variances only
- B. Equality of three or more population means
- C. Whether a single sample mean equals a constant
- D. Whether a population proportion equals 0.5

Part C — ANOVA requirements (Select all that apply)

Which conditions are listed in the lecture for one-way ANOVA?

- A. Populations are approximately normal
- B. Populations have the same variance (or standard deviation)
- C. Samples are simple random samples of quantitative data
- D. Samples are independent
- E. Variables must be categorical only
- F. Different samples are from populations categorized in only one way

Part D — Non-parametric tests (Short Answer)

State one advantage and one disadvantage of non-parametric tests given in the lecture.